

# Devansh Parapalli

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## Professional Summary

AI/ML engineer specializing in LLM inference, agentic systems and RAG pipelines. Delivered production systems at Cisco, automating support, audit and FinOps workflows that save 1,300+ person-hours monthly while triaging 700+ cases per month. Published in IJTE (2025). Certified GCP Professional Cloud Architect.

## Work Experience

**Software Development Engineer** Jul 2025 — Present  
Tata Consultancy Services Limited *Hyderabad, India*

- Client: Cisco** – Defined requirements, owned architecture, and drove cross-functional alignment across Cisco support, FinOps, and audit teams as the sole engineer across 3 concurrent production engagements.
  - Built CaseIQ end-to-end to triage and automate 700+ monthly support cases in under 30 seconds (down from ~24 business hours). Designed multi-intent detection, agentic tool-calling workflows and structured output delivery over a Kafka-backed event-driven ingestion pipeline with auto-scaling containerized runners.
  - Replaced manual payroll review (500+ person-hours/month) with an LLM harness that ingests PAAT records and Oracle GL accruals, parses email justifications, scores acceptability against policy, and tracks outstanding gaps – validated end-to-end with finance and audit stakeholders.
    - Prototyped an XGBoost-based transaction classifier – automating the upstream manual tagging workflow.
  - Built four async LLM-orchestrated FinOps pipelines (saving 800+ person-hours/month) spanning multi-source report generation across Snowflake/MongoDB/Oracle, a RAG knowledge chatbot, NL-to-query translation, and automated Excel delivery.
  - Architected orchestration per workload using custom LLM harnesses, MCP and FastMCP servers to expose internal data sources and tools as composable, reusable interfaces across pipelines.
- Internal: GenAI Center of Excellence** – Led technical architecture and client demos for 8 RFPs, contributing to engagements spanning GenAI enablement, cloud migration, and legacy modernization.
  - Architected and prototyped an on-premises vLLM + LiteLLM inference platform with centralized API gateway routing, PII detection, RBAC, and audit logging, enabling internal model serving without external API dependencies.

**GenAI & Backend Development Intern** Jun 2024 — Aug 2024  
PolymathAI *Nagpur, India*

- Cut per-video multi-modal AI processing time from **3 minutes to 15 seconds** by designing a parallelized inference pipeline across horizontally-scaled containers, processing 14,600+ videos in production.

## Projects

**AIKO: AI-powered Knowledge Organizer** ([aiko.parapalli.dev](https://aiko.parapalli.dev)) Jun 2024 — Present

- Achieved **Mean Reciprocal Rank 0.75 / nDCG@5 0.7** over a 100K-chunk technical knowledge corpus by implementing hybrid retrieval (semantic search + keyword fallback) with tuned chunking and embedding selection across multi-modal content.
- Sustained **p95 ingest latency of 3s** across a multi-modal document pipeline (PDFs, web content, multimedia) via a custom serialization layer with cloud-native and offline-capable deployment targets.
- Reduced LLM response latency by **25%** by replacing SSE with custom **WebSocket** duplex streaming for real-time token delivery.
- Built a custom orchestration framework for full control over prompt construction, retrieval, and tool-calling – replacing third-party abstractions with task-specific logic.

**ARA: AI-powered Research Assistant** ([ara.parapalli.dev](https://ara.parapalli.dev)) Jan 2024 — May 2024

- Built a RAG-based research workspace with live web retrieval and per-prompt source citation across FastAPI, SvelteKit, Supabase, and containerized LLM runners – predating mature RAG frameworks and requiring custom retrieval and orchestration logic.

## Education

**Government College of Engineering, Nagpur** Nagpur, India  
*B.Tech, Computer Science and Engineering* Dec 2021 — Jun 2025  
CGPA: 8.28/10.0

## Technical Skills

**AI/ML:** LLM Inference & Serving (vLLM, LiteLLM), Tool Calling, MCP, Agentic Systems, RAG, Hybrid Retrieval, XGBoost

**Languages:** Python, JavaScript/TypeScript, C/C++, SQL

**Frameworks & Libraries:** LangGraph, FastMCP, FastAPI, React.js/Next.js, Svelte

**Cloud & Infrastructure:** GCP (Certified Architect), AWS (EC2, S3, EKS), Docker, Kubernetes, OpenShift, Terraform

**Databases & Data Systems:** PostgreSQL w/ pgvector, Snowflake, MongoDB, Oracle, Spanner, Kafka

## Certifications & Achievements

- GCP Certifications:** Professional Cloud Architect (Dec 2025) | Cloud Digital Leader (Sep 2025) | Generative AI Leader (Sep 2025)
- Publication:** “AI-based Knowledge Organizer for diverse data formats” - IJTE, March 2025